LADA: Latent-Space Adversarial Diffusion Attack in Remote Sensing

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***Abstract*— Deep neural networks (DNNs) have achieved remarkable progress in remote sensing image (RSI) analysis, yet their vulnerability to subtle adversarial perturbations poses a critical threat to safety-critical applications such as environmental monitoring. While black-box attacks have garnered attention for their practicality, existing methods face a dilemma in RSI scenarios: restricted attacks often result in compromised image quality and limited stealthiness, whereas unrestricted attacks risk degrading transferability. To address this challenge, this article proposes the latent-space adversarial diffusion attack framework (LADA), which focuses on balancing stealthiness and transferability in adversarial attacks against RSI models. LADA employs a pre-trained diffusion model to map high-resolution RSIs into a low-dimensional latent space, enabling semantic-level perturbation optimization while avoiding pixel-wise explicit noise. Additionally, text prompts are automatically generated using a large multimodal model to guide adversarial sample synthesis, ensuring semantic consistency. To enhance perturbation search efficiency in the latent space, a hybrid strategy combining multiscale sampling and covariance matrix adaptation evolution strategy (CMA-ES) is introduced. Extensive experiments demonstrate that LADA achieves superior performance across multiple RSI datasets, model architectures, and defense mechanisms. This article establishes a benchmark for high-stealthiness and high-transferability adversarial attacks, advancing the secure deployment of DNNs in remote sensing applications.**

Received 3 April 2025; revised 27 May 2025 and 23 June 2025; accepted 5 July 2025. Date of publication 10 July 2025; date of current version 17 July 2025. This work was supported in part by China Postdoctoral Science Foundation under Grant 2024M762632, in part by the Natural Science Foundation of Shaanxi under Grant 2024JC-YBQN-0687, in part by the Social Science Foundation of Shaanxi under Grant 2024R027, in part by the Fundamental Research Funds of the Central Universities of China under Grant 245202301, in part by the National Natural Science Foundation of China under Grant 62103311, and in part by the Key Research and Development Program of Shaanxi under Grant 2025CY-YBXM-046. *(Corresponding author: Tao Zhan.)*

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Digital Object Identifier 10.1109/TGRS.2025.3587684

***Index Terms*— Adversarial attacks, diffusion model (DM), evolution search, multiscale sampling, remote sensing image (RSI).**

I. Introduction

DEEP neural networks (DNNs) have revolutionized remote sensing image (RSI) analysis, having spearheaded advancements in tasks such as classification, target detection, and scene interpretation through their capacity to automatically learn hierarchical feature representations [1], [2], [3]. However, the vulnerability of DNNs to imperceptible adversarial perturbations have been increasingly exposed, where subtle input modifications (e.g., via gradient-based methods like FGSM [4] or black-box transfer attacks [3], [5]) can induce catastrophic misclassifications, compromising decision-making in applications like environmental monitoring and disaster response [6]. Therefore, it is imperative to explore adversarial attack methods specific to the RSI fields to reveal the underlying vulnerabilities of DNNs [3], [7].

Adversarial attacks involve the intentional injection of imperceptible perturbations into input data to mislead machine learning models into producing erroneous outputs [4], [5]. These attacks are categorized into white-box attacks [4], where the adversary has full access to the architecture, parameters, and gradients of the victim model, and black-box attacks, which operate under no knowledge of the internal structure of the target model [5], [8]. In particular, black-box attacks have emerged as a focal point of research due to their practical relevance in real-world scenarios, where attackers typically lack complete model information [3], [7], [8]. The cross-model effectiveness of these transferable adversarial examples increases their stealth and adaptability, making them particularly concerning in security-sensitive applications like RSI [3], [7].

Black-box attacks can be divided into two major categories: restricted and unrestricted attacks. Restricted attacks employ perturbation constraints through ℓ_p-norm bound (e.g., ℓ_∞, ℓ_2) to maintain perceptual similarity between original and adversarial examples [3], [4], [5], [7], [8]. Unrestricted attacks [9], [10], [11], [12], [13], [14] operate without ℓ_p-norm constraints by modifying semantic image attributes through color manipulation [15] and texture modification [16]. These methods typically leverage generative architectures [9], [10], [11] or content-aware strategies [12], [17] to produce visually natural perturbations.

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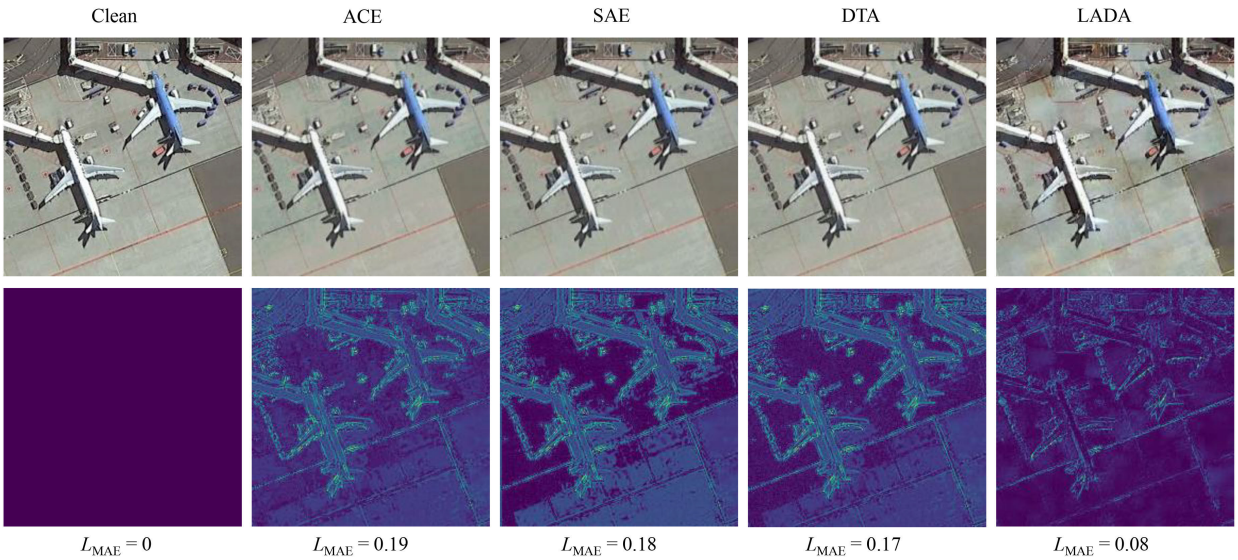


Fig. 1. MAE between some adversarial examples and clean images, where the first line represents the adversarial examples crafted by different attacks, the second line denotes the MAE visualization result between the adversarial examples and clean images, and L_{MAE} is the mean of MAE.

RSI scene classification primarily relies on spatial and textural features of the input [3], [18], [19], [20], [21], [22]. Existing adversarial attacks on RSI thus predominantly operate in the feature space [3], [18], [19], [21]. For instance, methods such as Mixup and Mixcut [3] directly manipulate the original feature distribution by blending or concatenating multiple input samples. FA [18] leverages shallow features extracted from a surrogate model to craft adversarial examples. As such, low-level features capture fundamental image characteristics that are often shared across different models. To further enhance cross-model transferability, DMFAA [22] introduces a distillation-based approach. While effective in improving transferability, it often produces unnatural noise due to severe corruption of original features. SFCoT [21] combines both spatial- and frequency-domain transformations to perturb the input and alter attention regions through strong interference. However, all these RSI attack methods are still constrained by an ℓ_p-norm bound to control perturbation magnitude. As a result, they fall under the category of restricted attacks, which often generate visually unnatural adversarial examples with noticeable noise artifacts.

As emphasized by Xue et al. [23], effective adversarial attacks must simultaneously satisfy two requirements: stealthiness and transferability. However, current methods [15], [16], [24], [25], [26] struggle to balance these objectives due to the NP-hard nature of optimizing perturbations in continuous input spaces under combined adversarial-effectiveness and perceptual-fidelity constraints. This fundamental challenge stems from the need to search high-dimensional spaces for minimal yet effective modifications while preserving semantic coherence.

In light of this, we propose a latent-space adversarial diffusion attack (LADA), a novel unrestricted adversarial attack framework specifically tailored for RSI. Leveraging diffusion models (DMs) [27], [28], [29], [30], LADA generates adversarial samples with significantly improved stealthiness and transferability. Unlike conventional pixel-level perturbations constrained by ℓ_p-norm [3], [4], [5], [7], [8], our framework maps RSIs to a low-dimensional latent space via an off-the-shelf pretrained DM [29], encoding high-level semantic information. This allows perturbations to be optimized at the semantic level rather than the pixel level, thus eliminating visible artifacts while preserving visual naturalness. To further enhance semantic consistency, we integrate text-guided diffusion by automatically generating descriptive prompts for each RSI using the Deepseek-R1[1] model. These prompts are encoded into text embeddings and fused with the latent space representation, ensuring adversarial samples retain original content and semantic integrity. To efficiently explore the latent space for optimal perturbations, we propose a hybrid optimization strategy combining multiscale sampling and the covariance matrix adaptation evolution strategy (CMA-ES) [31]. As shown in Fig. 1, the adversarial examples generated by our proposed LADA exhibit stronger stealthiness compared to state-of-the-art (SOTA) methods such as SAE [26], ACE [32], and DTA [33]. Our contributions to this work are summarized as follows.

1) We propose a novel unrestricted adversarial framework, LADA, for crafting adversarial RSI samples. LADA addresses the visibility limitations of pixel-level attacks by optimizing perturbations in latent space, achieving a unique balance between stealthiness and transferability for high-resolution RSIs.
2) By fusing text embeddings with latent representations, LADA ensures the generated adversarial samples retain original semantics, overcoming the semantic fragmentation common in existing unrestricted attacks. Furthermore, the multiscale iteration optimization-based CMA-ES method improves the efficiency of finding optimal perturbations (natural-looking and transferability) in the latent space.

[1]https://huggingface.co/deepseek-ai/DeepSeek-R1

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3) The extensive comparative experiments across multiple datasets, model architectures, and defense mechanisms have validated the superiority of LADA in RSI adversarial attacks. Further analysis using similarity metrics (e.g., peak signal-to-noise ratio (PSNR),[2] structural similarity index (SSIM) [34], learned perceptual image patch similarity (LPIPS) [35]) has demonstrated that LADA effectively balances stealthiness and transferability, achieving imperceptible perturbations while maintaining strong adversarial effectiveness against diverse defenses.

The remainder of this article is organized as follows. Section II provides an overview of related work in adversarial attacks and defense strategies. Section III formulates the problem and introduces necessary preliminaries. Section IV details the proposed LADA framework. Section V presents experimental results and performance analysis. Finally, Section VI concludes this article.

II. Related Work

A. Restricted Attack

Restricted attacks (ℓ_p norm-based adversarial attacks) aim to craft imperceptible perturbations constrained by specific distance metrics (e.g., ℓ_0, ℓ_2, ℓ_∞) [3], [4], [5], [7], [8]. The FGSM [4] uses ℓ_∞ norm to generate maximum single-step perturbations along gradient directions, enabling rapid attacks but suffering from low success rates against robust models. For ℓ_2 norm attacks, the Carlini & Wagner (C&W) method [8] minimizes perturbation magnitudes while ensuring misclassification through Curls iteration and Whey optimization, yielding high-success-rate attacks that bypass defensive distillation but require extensive hyperparameter tuning. Jacobian-based saliency map attack (JSMA) [36] employs ℓ_0 norm to sparsely modify critical features, generating human-imperceptible perturbations in image classification but becoming impractical in high-dimensional spaces due to exponential search complexity. However, existing studies [37], [38] demonstrate that while these constrained attacks achieve high success rates, the generated adversarial examples often deviate significantly from the distribution of natural images.

B. Unrestricted Attack

Unrestricted adversarial attacks generate perturbations unconstrained by ℓ_p-norms, focusing instead on manipulating image-level features like shape, texture, or color to evade detection [15], [16], [23], [26]. The texture-based approaches [16] alter global attributes like style or texture, though they frequently produce unnatural artifacts and exhibit weak adversarial robustness. Researchers then turned to color-based attacks [15], [26], which modify pixel values along specific dimensions to enhance photorealism. However, these approaches often sacrifice adaptability, as they rely on subjective heuristic criteria or restrictive loss functions to maintain visual plausibility. To address these limitations, recent works [12], [13] have combined adversarial loss with surrogate losses (e.g., content, style, smoothness) or generative models like GANs [11] to produce natural-looking adversarial examples. For instance, AdvCAM [39] enforces content and style preservation, while AdvArt [40] customizes adversarial patches with specific artistic styles. Despite improved imperceptibility, these methods struggle with loss function balancing, as overemphasizing realism often weakens adversarial strength. In contrast, the proposed LADA method searches for optimal perturbations in the latent space of DMs, preserving semantic and content integrity while achieving superior transferability across various victim models.

C. Adversarial Attacks in RSI

Adversarial attacks in RSI aim to deceive DNNs by introducing imperceptible perturbations to input data, impacting critical tasks, such as scene classification [3], [18], [19], [20], [21], object detection [7], [41], [42], and semantic segmentation [3], [43]. RSI scene classification models predominantly rely on spatial and textural features, making them vulnerable to feature-space manipulations. Early adversarial methods, such as Mixup and MixCut [3], directly alter the original feature distribution by blending or concatenating multiple input samples. These approaches perturb feature correlations but often lack transferability across models. To improve attack stealthiness, DeMPAA [19] deploys multiple localized adversarial patches on original images, achieving high attack success rates (ASRs). However, such patch-based methods introduce visible artifacts, compromising naturalness. For better cross-model transferability, DMFAA [22] adopts a distillation-based framework, though it tends to generate unnatural noise due to excessive feature corruption. SFCoT [21] combines spatial- and frequency-domain transformations to perturb inputs and shift model attention regions via aggressive interference. In RSI semantic segmentation, where images cover diverse objects and scenes, global adversarial perturbations are often inefficient. SASS [43] addresses this by selectively attacking only critical regions rather than the entire image, aligning with real-world scenarios where irrelevant areas need not be perturbed. However, most methods restrict perturbations via ℓ_p-norm bounds to ensure imperceptibility. This often results in unnatural noise patterns and visually detectable artifacts.

D. Adversarial Defense

Adversarial defense strategies aim to enhance the robustness of DNNs against adversarial attacks, primarily through input preprocessing defenses and adversarial training models [37], [38], [44]. Input preprocessing defenses [45], [46] involve techniques such as a high-level representation guided denoiser (HGD) [46], which utilizes high-level feature representations to guide the denoising process, effectively removing adversarial perturbations. Randomization and projection (R&P) methods [44] apply random transformations to input data, disrupting the structure of adversarial perturbations. DiffPure [37], [38] employs DMs for adversarial purification by adding noise to adversarial examples and then recovering the clean image through a reverse generative process. On the other hand, the adversarial training methods [47] incorporate adversarial examples during the training phase, enabling models to

[2]https://en.wikipedia.org/wiki/Peak_signal-to-noise_ratio

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learn to maintain performance even when faced with perturbed inputs. To verify the robustness of our proposed LADA method against adversarial defense techniques, we design comparative experiments in this article.

III. Preliminaries

A. Problem Formulation of Adversarial Attack

Given a victim model F parameterized by θ and a clean RSI input x with ground-truth label y, the problem formulation of adversarial attack with ℓ_p-norm is

$$\arg\max_{x_{\text{adv}}} \mathcal{L}(F_\theta(x_{\text{adv}}), y), \quad \text{s.t. } \|x_{\text{adv}} - x\|_p \leq \varrho \tag{1}$$

where $\mathcal{L}(\cdot)$ denotes a loss function, x_{adv} represents an adversarial example with a subtle perturbation, and the constraint is a ℓ_p-norm ball $\mathcal{B}_\varrho(x)$ with radius $\varrho > 0$. Thus, ℓ_p-norm-based methods aim to find an optimal perturbation δ from $\mathcal{B}_\varrho(x)$.

In contrast, unrestricted adversarial attacks craft the imperceptible perturbations without using ℓ_p-norm ball $\mathcal{B}_\epsilon(x)$, which suggests that the perturbations are natural [12], [13], [17]. However, the problem in (1) is impracticable to solve due the unknown θ. In light of this, we focus on the unrestricted adversarial attacks in a black-box setting and introduce a surrogate model F_ϕ to generate x_{adv} to fool the victim model F_θ. Thus, it is critical to exploit the transferability of F_ϕ and x_{adv} for boosting the ability to deceive various models across RSI domains. On the other hand, although the existing unrestricted attacks can improve the ASR, they tend to unrestrictedly change color, texture, and shape [15], [16], [24], [26]. As a result, the content of the clean input is significantly degraded, and the adversarial transferability is weakened. This insight highlights that directly modifying the original image content is inadvisable. Such aggressive perturbations cause catastrophic disruption to the structural integrity of the clean images and inherently lack reliable transferability across diverse models. To this end, we employ the inverse sampling capability of DMs, which enables the extraction of high-level semantic features through their denoising process.

B. Latent DMs

Latent DMs (LDMs) [28], [29], [30] are a class of generative models that operate in a low-dimensional latent space rather than a high-dimensional pixel space. A pretrained autoencoder (e.g., VQ-VAE [48], AutoencoderKL [29]) compresses input images into a compact latent representation that preserves semantic information while discarding perceptually irrelevant details. Mathematically, the encoder maps an image $x \in \mathbb{R}^{H\times W\times 3}$ in RGB space into a low-dimensional manifold in latent space, denoted as: $z = \mathcal{E}(x)$, $z \in \mathbb{R}^{h\times w\times c}$, where c is the channel dimension in latent space and $h \times w$ is the compressed spatial resolution. Specifically, $\mathcal{E}$ samples the input image x by a factor of $\gamma = H/h = W/w$. Then, LDMs use the decoder $\mathcal{D}$ to reconstruct the image from z, i.e., $\bar{x} = \mathcal{D}(z)$. The Kullback–Leibler (KL) divergence is applied during the autoencoder training to prevent the high-variance latent space originating from the random downsampling. During the latent diffusion process, DMs are applied to operate in the latent space z. For the forward process, an input image is gradually transformed into a Gaussian noise over T steps by

$$q(z_t|z_{t-1}) = \mathcal{N}(z_t; \sqrt{1-\beta_t}\cdot z_{t-1}, \beta_t \cdot \mathbf{I}) \tag{2}$$

where $z_0 = \mathcal{E}(x)$ and β_t is the noise schedule. We can directly obtain z_t from z_0 by

$$z_t = \sqrt{\bar{\alpha}_t}\cdot z_0 + \sqrt{1-\bar{\alpha}_t}\cdot \epsilon \tag{3}$$

where $\bar{\alpha}_t = \Pi_{s=1}^{t}(1-\beta_s)$ and $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. When $\bar{\alpha}_T$ is sufficiently close to 0, $z_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. From a variational theory, a larger T of the forward process tends to make z_t close to a standard Gaussian distribution. In the reverse/denoising process, a neural network parameterized with θ is used to predict noise, given by

$$p_\theta(z_{t-1}|z_t) = \mathcal{N}(z_{t-1}; \mu_\theta(z_t, t), \Sigma_\theta(z_t, t)) \tag{4}$$

where μ_θ and Σ_θ are the output of a model θ.

IV. Methodology

A. Framework Overview

The framework of LADA is shown in Fig. 2. Leveraging the latent space of DMs, we design a two-stage pipeline: first, we map input RSI into a compact latent representation using a pretrained LDM[3] to preserve critical spatial and spectral features while reducing computational complexity. To further enhance image generation quality and preserve semantic integrity, we incorporate text-guided DM techniques while integrating null-text inversion [30] to mitigate cumulative error propagation inherent in DMs. Specifically, we leverage the DeepSeek-R1 model to automatically generate text captions for RSIs, enabling precise semantic alignment between visual content and textual descriptions. Second, we inject imperceptible perturbations into the latent codes through a diffusion-based adversarial process, where a noise schedule and cross-attention mechanism dynamically adapt perturbations to evade detection while maintaining semantic plausibility. To maximize attack effectiveness, we employ the CMA-ES [31] to iteratively optimize latent perturbations δ, balancing stealthiness and model misclassification. By operating in the latent space, LADA achieves unrestricted attacks (free from pixel-wise constraints) while maintaining high transferability across different RSI models. Our framework further integrates multiscale sampling $\mathcal{M}(\cdot)$ to propagate adversarial signals across resolutions, ensuring robustness against spatial distortions common in remote sensing data.

B. Latent Space Mapping

Considering that RSIs not only contain rich surface and object information but also have high-resolution pixels, directly applying conventional DMs would significantly increase text-to-image generation time and the computational cost of gradient calculations. Additionally, explicit Markov chains require stepwise noise addition and heavily depend on the state of the previous step, making them

[3]https://huggingface.co/CompVis/stable-diffusion-v1-4

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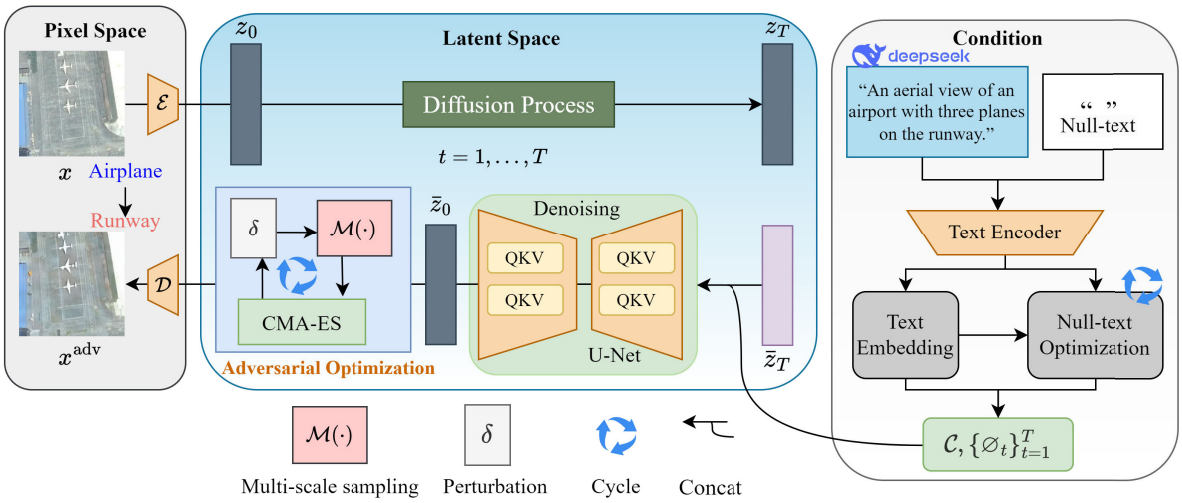


Fig. 2. Workflow of LADA: incorporating latent space mapping, condition concatenation, and latent variable optimization with multiscale sampling and CMA-ES iterative perturbation search.

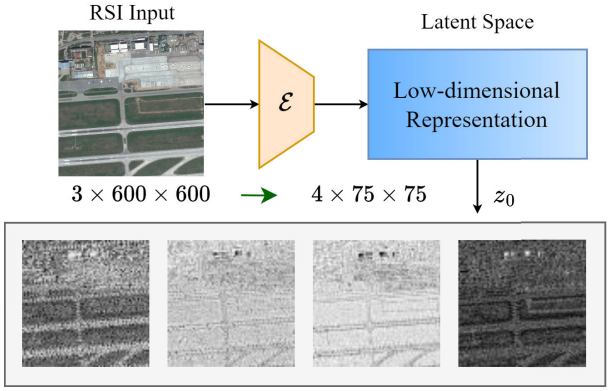


Fig. 3. RSI input with dimensions $3 \times 600 \times 600$ is mapped to a low-dimensional representation z_0 in the latent space with dimensions $4 \times 75 \times 75$, corresponding to a scale factor of $\gamma = 8$.

constrained by the total number of timesteps T. To address these challenges, we integrate image-perceptual compression and a non-Markovian diffusion-denoising process [28] into latent space mapping. Specifically, to obtain a perceptually rich low-dimensional representation, we employ a pretrained autoencoder [29] to provide a lower-dimensional latent space that is perceptually equivalent to the pixel space. As shown in Fig. 3, the low-dimensional representation in the latent space effectively captures the key information of the original RSI. Consequently, LDMs can perform the diffusion process within the latent space, allowing z_T to approximate a Gaussian distribution, similar to the diffusion process in conventional DMs. If $\mathcal{E}$ samples the x by a factor of γ, the computational complexity will be reduced at a rate of $\mathcal{O}(\gamma^2)$.

In addition, we adopt an implicit mapping that directly connects the initial latent variable z_0 with the noise z_t at timestep t. Following the non-Markovian forward process in DDIM [28], we can skip certain timesteps during the reverse sampling process to enhance the efficiency of RSI generation. Specifically, given an original sequence of timesteps $\{1, 2, \ldots, T\}$, DDIM selects a subsequence $\{t_1, t_2, \ldots, t_s\}$, allowing the forward process to be expressed as

$$z_{\tau_i} = \sqrt{\alpha_{\tau_i}} z_0 + \sqrt{1 - \alpha_{\tau_i}} \epsilon. \tag{5}$$

Due to the large interval between subsequence timesteps, DDIM does not need to generate $z_{t-1} \rightarrow z_t$ sequentially. Instead, it directly infers $z_{\tau_{i-1}}$ from z_{τ_i}. Similarly, the reverse process of DDIM is achieved through a deterministic mapping, formulated as follows:

$$z_{t-1} = \sqrt{\alpha_{t-1}} \left(\frac{z_t - \sqrt{1 - \alpha_t} \epsilon_\theta(z_t, t)}{\sqrt{\alpha_t}} \right) + \sqrt{1 - \alpha_{t-1} - \sigma_t^2} \cdot \epsilon_\theta(z_t, t) + \sigma_t \epsilon \tag{6}$$

where $\epsilon_\theta(z_t, t)$ represents the trained noise prediction network, σ_t is the noise scale parameter, which controls randomness, and $\epsilon \sim \mathcal{N}(0, I)$. The implicit definition of the forward process allows the reverse sampling to skip steps, accelerating the generation speed by 10 to 50 times.

C. Accurate Reconstruction With Null-Text

In adversarial attacks within the latent space, the absence of explicit constraints on content and semantics may cause the generated adversarial images to deviate significantly from the original clean input. To address this issue, it is essential to introduce a conditional mechanism to guide the image generation process [29]. To obtain semantic guidance for each RSI image, we first employ DeepSeek-R1 to generate descriptive textual prompts. Specifically, DeepSeek-R1 utilizes a vision encoder to extract high-dimensional visual features from the input, which are then projected through a Transformer-based vision-language mapping network. A fine-tuned prompt generator, built upon the GPT-2 architecture, subsequently produces concise textual descriptions that capture key visual attributes such as object recognition, scene composition, artistic style, and color distribution. To ensure semantic alignment with the original input, each prompt is limited to no more than 20 words, providing precise textual supervision for the diffusion.

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In this context, DMs inherently represent a conditional probability distribution $p(z|\mathcal{C})$, where $\mathcal{C}$ denotes the conditioning input–typically a textual description $\mathcal{P}$ that characterizes the image content. Text-guided diffusion models (e.g., DiffusionCLIP [49]) first encode the input prompt $\mathcal{P}$ via a pre-trained text encoder to obtain a semantic embedding, which then guides the denoising trajectory to align the generated image with the intended semantics. Specifically, the objective function for the diffusion process is given by

$$\min_{\theta} \; E_{z_0,\epsilon\sim N(0,I),t\sim\text{Uniform}(1,T)}\|\epsilon-\epsilon_\theta(z_t,t,\mathcal{C})\|_2^2$$

where $\mathcal{C}=\phi(\mathcal{P})$ is the text embedding, $\phi(\cdot)$ is the embedding functions, and ϵ_θ uses z_t and $\mathcal{C}$ to predict noise.

Remark 1: The training objective of the DMs minimizes the mean squared error (mse) between the predicted noise and the actual injected noise. During training, the model learns a noise estimator ϵ_θ that takes the noisy input z_t, timestep t, and a semantic embedding $\mathcal{C}=\phi(\mathcal{P})$ derived from the text prompt $\mathcal{P}$. This design enables the model to incorporate text-based semantic information into the denoising process, ensuring that the generated output remains faithful to the input description.

To accurately reconstruct the RSI input, (6) is reused and rewritten as

$$z_{t-1}=\sqrt{\frac{\alpha_{t-1}}{\alpha_t}}z_t+\left(\sqrt{1-\alpha_{t-1}}-\sqrt{\frac{\alpha_{t-1}(1-\alpha_t)}{\alpha_t}}\right)\times\epsilon_\theta(z_t,t,\mathcal{C}) \quad (7)$$

where $\sigma_t=0$ for a deterministic sampling. However, as presented in [30], the reconstructed output may be induced by $\mathcal{C}$ by a hard text-guided DMs [49], [50]. Thus, we can use a popular classifier-free guidance method [30]. To be concrete, let $\varnothing=\phi(\text{“ ”})$ be a null-text embedding of and $\zeta\geq 0$ control the guidance scale, the classifier-free guidance method for noise prediction is given by

$$\tilde{\epsilon}_\theta(z_t,t,\mathcal{C},\varnothing)=\zeta\epsilon_\theta(z_t,t,\mathcal{C})+(1-\zeta)\epsilon_\theta(z_t,t,\varnothing). \quad (8)$$

Since the noise prediction is conducted unconditionally, prediction errors can accumulate at each step of the inverse diffusion process. This accumulated error is further exacerbated by the scaling factor ζ, which amplifies deviations over successive iterations. As a result, the generated image may gradually drift away from the intended reconstruction, leading to degradation in visual fidelity and semantic consistency. Thus, we employ the null-text optimization technique proposed in [30], which treats the null-text embedding $\varnothing$ as a variable. In view of this, we need to iteratively find an optimal $\varnothing_t$ in each timestamp $t=T,\ldots,1$ to minimize the following objective:

$$\min_{\varnothing_t} \; \left\|z^*_{t-1}-z_{t-1}(\bar{z}_t,\varnothing_t,\mathcal{C})\right\|_2^2 \quad (9)$$

where $z^*_t\in\{z^*_T,\ldots,z^*_0\}$ is a sequence of noisy samples in latent space generated by the DDIM inversion process with $\zeta=1$ and $z^*_0=z_0$. z_{t-1} is given by

$$z_{t-1}=\sqrt{\frac{\alpha_{t-1}}{\alpha_t}}\bar{z}_t+\hat{\alpha}\tilde{\epsilon}_\theta(z_t,t,C,\varnothing_t). \quad (10)$$

We can use $z_{t-1}(\bar{z}_t,\varnothing_t,\mathcal{C})$ to denote $\bar{z}_{t-1}$. By leveraging the optimized null-text embedding $\{\varnothing_t\}_{t=1}^T$ and the diffused noise $\bar{z}_T$, we can achieve precise editing and generation of RSIs. This addresses the lack of control inherent in directly using DMs to generate adversarial images. In other words, our approach enables fine-grained control over the generated adversarial images based on RSI-specific prompts, allowing for precise manipulation of the mask and style references.

D. Multiscale Iterative Optimization Based on CMA-ES

After obtaining the denoised low-dimensional latent variable $\bar{z}_0$, we apply adversarial perturbation to it. The key motivation is that $\bar{z}_0$ encapsulates both the semantic and content information of the original image. By directly adding a subtle perturbation to $\bar{z}_0$, we can precisely control the perturbation magnitude while avoiding interference with the DM process. This approach reduces unnecessary computational overhead and helps preserve the original semantic and content integrity. Formally, let $\hat{z}_0=\bar{z}_0+\delta$ be the adversarial latent variable, the problem formulation is defined as

$$\arg\max_{\hat{z}_0} \; \mathcal{L}(F_\theta(\mathcal{D}(\hat{z}_0)),y), \quad \text{s.t. } \|\delta\|_\infty\leq\varrho. \quad (11)$$

Optimizing solely for maximal attack strength can result in adversarial images that deviate significantly from the distribution of the clean input. Our objective is to achieve a high ASR while maintaining perceptual similarity to the original image, ensuring that the perturbations remain imperceptible to the human eye. To this end, we reformulate the objective function as follows:

$$\arg\max_{\hat{z}_0} \; \{\mathcal{L}_{\text{CE}}(F_\theta(\mathcal{D}(\hat{z}_0),y)-\lambda\mathcal{L}_{\text{mse}}(\mathcal{D}(\hat{z}_0),x)\} \quad (12)$$

where $\lambda\geq 0$ is a hyperparameter to control the attack strength, $\mathcal{L}_{\text{CE}}$ and $\mathcal{L}_{\text{mse}}$ are cross-entropy loss and mse loss. $\mathcal{D}$ is a decoder to reconstruct the RSI from $\hat{z}_0$. Unlike prior approaches [12], [13], [17], [23], we refrain from directly optimizing the denoised latent representation $\bar{z}_0$. This decision is motivated by the inherent computational complexity associated with searching for optimal perturbations, even when operating within a lower dimensional latent space. An intuitive solution of further downsampling the latent representation to reduce dimensionality comes at the cost of significant semantic degradation, potentially leading to suboptimal perturbations that remain confined to local minima. To this end, we propose a multiscale sampling strategy that facilitates a more effective perturbation search. By leveraging hierarchical representations at multiple scales, LADA preserves essential semantic information while substantially mitigating computational overhead. This approach ensures that adversarial perturbations remain both semantically coherent and computationally efficient.

Fig. 4 shows the principle of the multiscale iterative optimization strategy based on CMA-ES. After DDIM latent mapping, a potential representation $\bar{z}_0$ with a size of $C\times H\times W$ is obtained, where C represents the number of channels, H denotes the height, and W is the width. Subsequently, we will adopt CMA-ES to search for an optimal latent perturbation at different scales and add it to $\bar{z}_0$ in turn to achieve attacks in the

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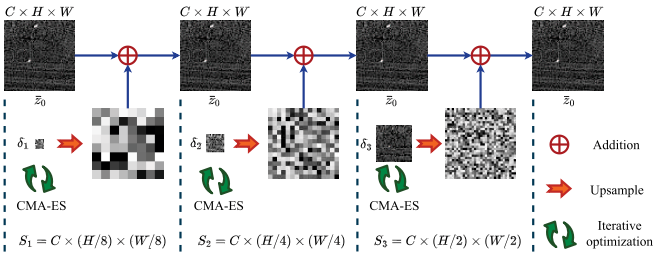


Fig. 4. Schematic of multi-scale iterative optimization based on CMA-ES, where $\bar{z}_0$, δ, S, C, H, and W denote latent representation, perturbation, optimizing scale, the number of channels, height, and width of the feature map, respectively.

latent space. Specifically, the perturbation with the dimension of $C \times (H/f) \times (W/f)$ at different scales is initialized as the optimization variable of CMA-ES, where f is the sample factor. In Fig. 4, δ_1, δ_2, and δ_3 represent the perturbations obtained when f is 8, 4, and 2, respectively. Therefore, the above optimization process is as follows:

$$\bar{z}_k = \bar{z}_{k-1} + \mathcal{I}_k(\Psi_k(\delta_k)) \tag{13}$$

where $\bar{z}_k$, $\mathcal{I}_k$, Ψ_k, and δ_k represent the latent representation, the upsampling operation, the CMA-ES optimizer, and the perturbation at the k-th scale, respectively. The objective function of the CMA-ES optimizer is as follows:

$$\arg\max_{\delta_k} \{\mathcal{L}_{\text{CE}}(F_\theta(\mathcal{D}(\bar{z}_{k-1} + \mathcal{I}_k(\delta_k)), y) - \lambda \mathcal{L}_{\text{mse}}(\mathcal{D}(\bar{z}_{k-1} + \mathcal{I}_k(\delta_k)), x)\}. \tag{14}$$

The first item guides the surrogate model to misclassify, achieving adversarial attacks. The second item makes the adversarial samples as similar as possible to the clean images to ensure the stealthiness of the attack. These perturbations at different scales will affect $\bar{z}_0$ at different pixel levels, which can improve the attack efficiency and are suitable for processing high-resolution RSIs containing rich surface and object information. Moreover, perturbations at different scales are added to $\bar{z}_0$, and the influence of existing perturbations will be further considered in the subsequent scale optimization process, which can give full play to the interaction of perturbations at multiple scales, thus ensuring the final attack effect.

At a single scale, the process of CMA-ES optimizing perturbation is shown in Algorithm 1. A random perturbation is initialized according to the current scale S, and its distribution mean m and covariance matrix C are calculated and assigned, where $\mathbf{I}$ represents the unit matrix (lines 1–3). The process of CMA-ES optimization mainly includes three parts: the generation of new solutions, the calculation of fitness, and the update of parameters. First, new candidate solutions are generated by sampling from a multidimensional normal distribution determined by m, ξ, and C. Second, the fitness Fit of these candidate solutions is calculated based on the objective function (14). However, the solutions of traditional CMA-ES are 1-D vectors, which are inconsistent with the dimension of the latent representation. Therefore, the dimension of these candidate solutions will be adjusted to S to facilitate the calculation of the objective function (lines 6–8). Thirdly,

Algorithm 1 Perturbation Optimization Based on CMA-ES

Input: $\bar{z}_{k-1}$ (latent representation at previous scale), $\xi > 0$ (optimization step size), $maxIter$ (maximum number of iterations), S (current scale)
Output: δ_k (optimized perturbation)

1: $\delta_k \leftarrow$ Initialize the perturbation according to S;
2: $m \leftarrow \text{mean}(\delta_k)$;
3: $C \leftarrow \mathbf{I}$;
4: **for** $i \leftarrow 1$ to $maxIter$ **do**
5: **for** $j \leftarrow 1$ to λ **do**
6: $X_j \sim m + \xi \cdot \mathcal{N}(0, C)$;
7: $\hat{X}_j \leftarrow$ Adjust the shape of X_j;
8: $Fit_j \leftarrow$ Calculate fitness based on the Eq. (14);
9: **end for**
10: Sort X according to Fit;
11: $m \leftarrow \sum_{q=1}^{\mu} \omega_q X_{q:\lambda}$;
12: Update step size ξ and covariance matrix C;
13: **end for**
14: $\delta_k \leftarrow$ Select X with the best Fit;
15: $\delta_k \leftarrow$ Adjust the shape of δ_k;
16: $\delta_k \leftarrow \mathcal{I}_k(\delta_k)$.

all candidate solutions are sorted according to their fitness, and parameters m, ξ, and C are updated according to the top λ solutions, where λ and ω represent population size and recombination weight (lines 10–12) [51], respectively. After the maximum number of iterations is reached, the solution with the best fitness is adjusted in dimension and upsampled to obtain the optimal perturbation at the current scale (lines 14–16).

V. Experiments

A. Experimental Setup

1) *Dataset:* The aerial image dataset (AID) [52], the RSI classification dataset created by Northwestern Polytechnical University (NWPU-RESISC) [53], and RSSCN7 [54] corrected by Wuhan University are used to verify the performance of the proposed method. AID [52] contains 10 000 images with a resolution of 600 × 600 pixels, and they can be divided into 30 classes. NWPU-RESISC [53] has 31 500 images at a resolution of 256 × 256 pixels, which are categorized into 45 classes. RSSCN7 [54] contains 2800 images measuring 400 × 400 pixels, which are from seven typical scene classes. In the experiment, the images in the above three datasets will be resized to 224 × 224 pixels.
2) *Classification Model:* To evaluate the adversarial transferability, different pretrained network structures are adopted, including convolutional neural networks (CNNs) and Transformers. For CNNs, we utilize Alexnet [55], ResNet-18 (Res-18) [56], DenseNet121 (Dense-121) [57], RegNetX-400MF (Reg-400) [58], VGG-19 [59], ResNet-50 (Res-50) [56], ResNet-101 (Res-101) [56], ResNeXt-50 (ResN-50) [60], RegNetX-8GF (Reg-8GF) [58], and RegNetX-16GF (Reg-16GF) [58]. For Transformers, we consider ViT-B/16 (ViT-B)

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TABLE I
ASR(%) OF NINE BLACK-BOX ATTACK METHODS ON THE AID DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Surrogate Models	Methods	Target Models							
		CNNs						Transformers	
		VGG-19	Res-50	Res-101	ResN-50	Reg-8GF	Reg-16GF	ViT-B	MobViT-s
Alexnet	FGSM	55.30±0.00	60.50±0.00	74.20±0.00	44.40±0.00	42.90±0.00	51.33±0.06	30.90±0.00	47.00±0.00
	I-FGSM	57.97±0.12	61.30±0.10	74.73±0.06	45.50±0.20	44.73±0.31	52.20±0.17	31.70±0.00	49.67±0.12
	SAE	60.03±0.12	65.87±0.12	79.97±0.15	47.87±0.06	47.90±0.53	54.60±0.10	35.40±0.70	52.63±0.50
	ACE	58.50±0.00	64.60±0.00	79.10±0.00	47.00±0.00	45.90±0.00	53.70±0.00	34.40±0.00	50.20±0.00
	AI-FGTM	**61.30±0.00**	66.20±0.00	80.73±0.06	49.00±0.00	49.10±0.00	53.90±0.00	**37.00±0.00**	**57.10±0.00**
	DTA	59.70±0.00	66.03±0.06	80.00±0.00	48.13±0.06	48.20±0.00	54.10±0.10	35.23±0.06	53.43±0.06
	BFA	59.97±0.06	65.70±0.10	80.03±0.06	48.03±0.12	47.63±0.12	54.40±0.10	35.07±0.06	52.27±0.15
	DeCoWA	60.07±0.12	65.97±0.06	79.93±0.06	48.37±0.12	48.67±0.23	54.97±0.23	35.77±0.12	53.07±0.15
	SFCoT	56.40±0.10	60.60±0.12	74.90±0.06	44.90±0.12	43.70±0.10	51.70±0.20	32.00±0.02	48.60±0.10
	LADA	60.67±0.85	**69.20±1.04**	**82.37±0.47**	**49.33±0.75**	**56.07±0.97**	**60.20±0.35**	36.77±0.35	55.87±1.00
Resnet-18	FGSM	55.20±0.00	60.43±0.06	74.23±0.06	44.40±0.00	42.90±0.00	51.30±0.00	30.80±0.00	46.93±0.06
	I-FGSM	58.67±0.06	60.10±0.26	76.00±0.10	45.50±0.20	46.50±0.36	52.40±0.26	31.73±0.06	50.40±0.46
	SAE	59.93±0.61	65.60±0.35	79.87±0.21	47.93±0.74	47.80±0.36	54.30±0.66	35.53±0.25	52.10±0.36
	ACE	58.10±0.00	63.70±0.00	78.03±0.06	46.20±0.00	45.60±0.00	52.57±0.06	34.30±0.00	49.80±0.00
	AI-FGTM	61.97±0.06	67.13±0.12	81.97±0.12	49.60±0.00	51.70±0.00	55.07±0.12	36.00±0.10	56.77±0.06
	DTA	60.03±0.06	66.20±0.00	80.10±0.00	48.27±0.15	48.03±0.06	54.23±0.15	35.10±0.10	52.57±0.15
	BFA	60.30±0.00	66.83±0.15	80.40±0.10	50.03±0.06	49.80±0.10	54.63±0.21	35.03±0.12	53.80±0.20
	DeCoWA	60.50±0.10	67.13±0.12	80.63±0.06	49.23±0.23	50.13±0.38	55.40±0.17	35.87±0.12	53.83±0.06
	SFCoT	56.50±0.22	61.40±0.20	74.90±0.12	45.20±0.18	43.70±0.30	52.20±0.20	31.00±0.22	47.80±0.30
	LADA	**62.60±0.70**	**75.57±0.25**	**85.63±0.93**	**56.40±0.80**	**61.80±0.56**	**63.90±0.92**	**38.37±0.35**	**59.70±0.79**
Densenet121	FGSM	55.20±0.00	60.43±0.06	74.20±0.00	44.47±0.06	42.90±0.00	51.30±0.00	30.70±0.00	46.90±0.00
	I-FGSM	57.50±0.44	60.40±0.26	75.47±0.15	44.83±0.15	44.83±0.06	52.77±0.06	31.50±0.20	49.10±0.20
	SAE	59.87±0.15	65.73±0.15	79.63±0.40	47.77±0.46	48.00±0.30	54.10±0.20	35.37±0.59	52.37±0.40
	ACE	58.50±0.00	63.10±0.00	78.50±0.00	46.13±0.06	46.30±0.00	53.00±0.00	34.00±0.00	50.70±0.00
	AI-FGTM	62.17±0.06	66.80±0.10	81.07±0.06	49.17±0.06	50.70±0.10	54.83±0.06	35.37±0.06	55.53±0.06
	DTA	59.87±0.06	66.43±0.06	80.17±0.06	48.77±0.12	48.83±0.06	54.67±0.23	34.97±0.12	52.40±0.00
	BFA	60.60±0.10	66.80±0.10	80.77±0.15	49.47±0.32	50.20±0.20	55.47±0.32	35.20±0.20	53.93±0.21
	DeCoWA	60.73±0.12	66.93±0.29	81.00±0.17	49.53±0.06	50.20±0.17	55.83±0.06	35.60±0.10	53.90±0.20
	SFCoT	56.50±0.02	60.70±0.18	74.90±0.10	45.10±0.20	44.50±0.22	52.00±0.22	30.90±0.22	47.40±0.27
	LADA	**65.13±0.25**	**72.63±0.29**	**83.70±0.36**	**51.83±0.75**	**57.57±0.38**	**60.70±0.87**	**36.27±0.68**	**58.07±0.45**
Regnet_x_400mf	FGSM	55.30±0.00	60.40±0.00	74.20±0.00	44.40±0.00	42.90±0.00	51.30±0.00	30.70±0.00	47.00±0.00
	I-FGSM	57.07±0.06	60.20±0.35	75.33±0.15	44.50±0.10	44.60±0.36	51.13±0.15	31.10±0.10	48.43±0.15
	SAE	60.37±0.23	65.80±0.20	79.73±0.32	47.87±0.06	47.73±0.32	54.67±0.25	35.33±0.35	51.93±0.57
	ACE	58.23±0.06	64.10±0.00	78.00±0.00	46.03±0.06	46.40±0.10	52.57±0.06	34.20±0.00	49.43±0.06
	AI-FGTM	60.20±0.00	65.33±0.06	80.40±0.00	47.50±0.00	49.73±0.12	53.90±0.00	35.23±0.06	54.77±0.06
	DTA	59.43±0.06	65.33±0.12	79.60±0.00	47.33±0.06	47.83±0.06	53.57±0.12	34.83±0.06	51.57±0.06
	BFA	59.30±0.00	65.70±0.00	79.50±0.00	47.60±0.00	48.50±0.00	53.80±0.00	34.60±0.00	51.70±0.00
	DeCoWA	59.77±0.06	66.40±0.20	80.03±0.15	48.23±0.31	48.93±0.12	54.23±0.21	35.83±0.12	53.37±0.31
	SFCoT	56.02±0.02	61.18±0.16	75.43±0.12	45.59±0.22	45.19±0.20	52.30±0.20	31.30±0.12	48.60±0.25
	LADA	**65.50±0.26**	**71.17±0.60**	**83.43±0.65**	**51.47±0.57**	**58.10±0.36**	**61.43±0.45**	**36.13±0.55**	**57.93±0.40**

[61] and MobileViT (MobViT-s) [62]. In these models, the first four models are utilized as surrogate models for black-box attacks, and the other models are employed as victim models.

3) *Comparison Methods and Hyperparameters:* Nine SOTA attack methods are utilized as competitors to verify the performance of LADA, which are FGSM [4], I-FGSM [5], SAE [26], ACE [32], AI-FGSM [63], DTA [33], BFA [64], DeCoWA [65], and SFCoT [21]. SFCoT enhances adversarial transferability for remote sensing recognition by jointly transforming image components across spatial and frequency domains to reduce model-specific overfitting. The hyperparameters for these methods are set according to the optimal configurations reported in the original papers. In the process of generating adversarial examples, the perturbation constraints, step size, and the number of iterations are set to 0.1, 1, and 5, respectively. In LADA, ξ and maxIter are set to 0.1 and 50, respectively. Moreover, LADA optimizes perturbations at three different scales, and the corresponding sampling factors f are 8, 4, and 2, respectively.

4) *Evaluation Metric:* ASR [3] is adopted to evaluate the performance of black-box attacks, which is defined as follows: ASR $= n_{\text{error}}/n_{\text{total}}$, where n_{error} and n_{total} denotes the number of misclassified samples and the total number of samples. To assess the perceptual quality and semantic fidelity of adversarial examples, we employ the following three metrics. 1) *SSIM [34]:* Measures structural similarity between the adversarial and original images, reflecting human perceptual quality. 2) *PSNR:* Evaluates pixel-level distortion. 3) *LPIPS [35]:* Quantifies perceptual differences using deep network features.

5) *Implementation:* All experiments are implemented using the PyTorch framework and are performed on

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TABLE II
ASR(%) OF NINE BLACK-BOX ATTACK METHODS ON THE NWPU-RESISC DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Surrogate Models	Methods	Target Models							
		CNNs						Transformers	
		VGG-19	Res-50	Res-101	ResN-50	Reg-8GF	Reg-16GF	ViT-B	MobViT-s
Alexnet	FGSM	40.57±0.06	43.77±0.06	59.07±0.06	40.53±0.06	31.17±0.06	44.77±0.06	56.00±0.00	43.40±0.00
	I-FGSM	41.57±0.15	46.00±0.10	59.67±0.12	41.43±0.15	32.53±0.15	45.57±0.25	56.37±0.06	45.17±0.21
	SAE	43.27±0.15	47.30±0.44	62.67±0.06	44.53±0.32	33.40±0.36	49.80±0.20	60.00±0.26	47.30±0.70
	ACE	44.60±0.00	48.00±0.00	64.30±0.00	44.80±0.00	33.30±0.00	50.30±0.00	61.50±0.00	47.70±0.00
	AI-FGTM	45.30±0.00	50.70±0.00	65.00±0.00	46.80±0.00	35.80±0.00	50.90±0.00	62.60±0.00	48.90±0.00
	DTA	44.33±0.06	48.83±0.15	63.57±0.06	46.10±0.00	34.03±0.06	50.70±0.00	61.80±0.00	47.90±0.10
	BFA	44.80±0.10	49.33±0.12	64.40±0.00	46.23±0.15	34.00±0.10	50.47±0.06	61.37±0.06	47.80±0.00
	DeCoWA	44.97±0.12	49.63±0.15	64.23±0.15	46.67±0.15	34.27±0.12	51.00±0.17	62.23±0.06	48.43±0.06
	SFCoT	41.10±0.10	45.70±0.19	59.90±0.10	42.70±0.16	32.60±0.20	46.10±0.18	56.40±0.16	44.30±0.12
	LADA	**54.04±6.83**	**59.00±5.59**	**66.30±1.57**	**55.21±5.90**	**45.45±5.08**	**55.57±3.87**	**65.63±7.94**	**56.17±4.93**
Resnet-18	FGSM	40.53±0.06	43.70±0.00	59.00±0.00	40.50±0.00	31.13±0.06	44.70±0.00	56.00±0.00	43.37±0.06
	I-FGSM	41.33±0.12	45.37±0.42	60.27±0.06	42.23±0.23	31.97±0.21	45.73±0.15	56.27±0.15	44.67±0.50
	SAE	43.07±0.25	47.43±0.15	62.43±0.31	44.27±0.42	33.87±0.35	50.30±0.66	59.60±0.20	47.57±0.21
	ACE	44.17±0.06	48.80±0.00	62.90±0.00	44.90±0.00	33.30±0.00	50.20±0.00	61.90±0.00	47.70±0.00
	AI-FGTM	44.67±0.06	52.30±0.00	65.83±0.06	48.13±0.06	36.07±0.06	51.20±0.00	61.50±0.00	49.23±0.06
	DTA	44.23±0.06	49.87±0.15	65.10±0.00	46.63±0.06	34.53±0.06	50.57±0.06	61.47±0.06	48.00±0.00
	BFA	44.77±0.06	51.47±0.15	65.33±0.12	47.47±0.06	35.17±0.15	50.77±0.06	61.30±0.00	48.63±0.32
	DeCoWA	45.27±0.06	51.50±0.20	65.83±0.15	47.20±0.10	35.57±0.21	51.57±0.15	61.63±0.06	49.30±0.10
	SFCoT	41.20±0.20	45.50±0.18	59.90±0.30	42.70±0.14	32.20±0.15	45.09±0.18	56.10±0.16	44.30±0.29
	LADA	**56.21±4.66**	**63.41±5.07**	**69.42±1.38**	**59.03±4.94**	**50.37±5.78**	**58.71±1.09**	**64.87±3.70**	**57.56±2.51**
Densenet121	FGSM	40.53±0.06	43.73±0.06	59.00±0.00	40.60±0.00	31.10±0.00	44.70±0.00	56.00±0.00	43.27±0.06
	I-FGSM	40.93±0.12	45.17±0.29	59.27±0.15	42.13±0.15	31.53±0.21	45.47±0.31	55.93±0.12	44.30±0.17
	SAE	42.90±0.20	47.13±0.21	62.47±0.06	44.23±0.40	33.60±0.30	49.93±0.50	59.70±0.10	47.33±0.06
	ACE	43.50±0.00	48.60±0.00	63.10±0.00	44.83±0.06	32.50±0.00	50.10±0.00	61.10±0.00	48.30±0.00
	AI-FGTM	45.60±0.00	52.77±0.06	66.00±0.00	49.00±0.00	36.07±0.06	52.50±0.00	61.47±0.15	50.57±0.06
	DTA	45.00±0.10	50.30±0.00	64.83±0.06	47.17±0.06	33.87±0.06	51.10±0.00	61.00±0.00	48.23±0.06
	BFA	45.03±0.23	51.60±0.10	65.77±0.21	48.13±0.15	35.53±0.21	51.67±0.06	61.37±0.12	49.57±0.25
	DeCoWA	45.63±0.12	51.80±0.20	66.00±0.10	48.20±0.10	35.87±0.06	51.83±0.06	61.47±0.12	50.17±0.21
	SFCoT	40.80±0.20	45.10±0.16	59.10±0.22	42.50±0.18	32.92±0.24	45.40±0.18	56.00±0.16	44.50±0.26
	LADA	**52.40±8.33**	**65.63±7.60**	**68.13±6.42**	**64.20±8.21**	**48.25±8.50**	**59.50±6.38**	**65.78±9.33**	**55.54±9.75**
Regnet_x_400mf	FGSM	40.50±0.00	43.60±0.00	59.03±0.06	40.50±0.00	31.10±0.00	44.73±0.06	56.00±0.00	43.27±0.06
	I-FGSM	40.87±0.06	44.33±0.06	59.30±0.10	41.07±0.06	31.43±0.29	45.23±0.06	56.03±0.06	43.47±0.06
	SAE	42.77±0.06	47.03±0.23	62.43±0.29	44.30±0.36	33.70±0.17	50.10±0.00	60.00±0.36	48.07±0.32
	ACE	43.90±0.00	47.93±0.06	62.80±0.00	44.60±0.00	33.70±0.00	49.80±0.00	60.60±0.00	47.33±0.06
	AI-FGTM	44.57±0.06	51.10±0.00	64.50±0.00	46.50±0.00	35.20±0.00	51.00±0.00	61.50±0.00	49.03±0.06
	DTA	44.37±0.06	48.80±0.10	63.97±0.06	45.67±0.06	33.90±0.10	50.27±0.06	61.03±0.12	47.83±0.06
	BFA	44.10±0.00	48.80±0.00	63.90±0.00	45.40±0.00	33.50±0.00	50.17±0.06	61.20±0.00	47.80±0.00
	DeCoWA	44.73±0.06	50.47±0.15	65.07±0.12	46.67±0.15	35.23±0.12	51.23±0.06	61.83±0.21	48.67±0.12
	SFCoT	40.90±0.20	44.80±0.25	59.30±0.21	42.13±0.21	32.30±0.22	45.80±0.18	56.00±0.16	43.89±0.22
	LADA	**52.93±3.57**	**59.08±4.96**	**66.26±1.01**	**57.43±6.99**	**48.06±5.73**	**57.43±2.31**	**63.61±3.80**	**56.85±3.72**

a computer with NVIDIA RTX 3090Ti GPU of 24 GB memory. Given the massive scale of DeepSeek-R1 (with 671 billion parameters), deploying the model locally for image-to-text generation is impractical due to prohibitive computational and storage requirements. To address this, we adopt a cloud-based solution by accessing DeepSeek-R1 via an application programming interface (API), leveraging remote computation to efficiently handle the image understanding task. On the client side, input images are encoded into binary streams to facilitate efficient transmission and compatibility with the API interface. The input prompt is set as: *Please describe the image in a simple sentence of less than 20 words*.

B. Comparisons With SOTA Methods

1) *Quantitative Results:* A comprehensive evaluation of LADA and nine baseline methods is conducted on the AID, NWPU-RESISC, and RSSCN7 datasets. The average ASR and corresponding variance across five random trials are presented in Tables I–III. From the results, the adversarial samples generated by LADA can mislead most models and get higher ASR, which is superior to the other nine SOTA methods. Specifically, FGSM [4] and I-FGSM [5] only use simple linear adversarial perturbations, and most of their ASRs are less than 50%, showing poor performance. AI-FGTM [63] adopts the tanh function instead of the sign function to avoid the loss of gradient and utilizes an increased step size to add perturbations, which improves the ASR by about 5%. As a direction-tuning attack, DTA [33] reduces the angle between the actual update direction and the steepest update direction by adding an internal cyclic sampling module, achieving ASR similar to AI-FGTM. The relatively low ASR of SFCoT can be attributed to its reliance on larger perturbation magnitudes to generate effective adversarial examples. SAE and ACE attack the color

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TABLE III
ASR(%) OF NINE BLACK-BOX ATTACK METHODS ON THE RSSCN7 DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Surrogate Models	Methods	Target Models							
		CNNs						Transformers	
		VGG-19	Res-50	Res-101	ResN-50	Reg-8GF	Reg-16GF	ViT-B	MobViT-s
Alexnet	FGSM	45.36±0.00	67.32±0.00	48.57±0.00	38.51±0.10	45.71±0.00	47.32±0.00	33.10±0.10	41.43±0.00
	I-FGSM	46.61±0.36	67.62±0.21	48.93±0.64	40.54±0.18	47.38±0.41	46.79±0.47	34.23±0.10	43.57±0.62
	SAE	47.86±0.64	70.06±0.10	51.07±0.18	38.93±0.54	48.39±0.00	47.86±0.18	35.12±0.37	42.26±0.63
	ACE	47.14±0.00	69.46±0.00	50.18±0.00	39.46±0.00	48.04±0.00	46.43±0.00	34.82±0.00	42.32±0.00
	AI-FGTM	48.45±0.10	70.36±0.00	50.95±0.10	38.75±0.00	48.39±0.00	47.50±0.00	35.71±0.00	42.50±0.00
	DTA	47.86±0.00	70.18±0.00	50.71±0.00	39.29±0.00	48.51±0.10	46.96±0.00	35.48±0.10	42.62±0.10
	BFA	48.21±0.00	70.18±0.00	51.10±0.10	40.30±0.10	48.45±0.10	47.32±0.00	36.13±0.27	43.10±0.10
	DeCoWA	48.33±0.10	70.06±0.10	**51.19±0.10**	40.06±0.21	49.05±0.21	47.20±0.10	36.13±0.37	42.98±0.10
	SFCoT	46.61±0.16	68.03±0.20	49.11±0.12	38.93±0.24	46.07±0.22	47.32±0.20	33.40±0.29	41.96±0.30
	LADA	**51.44±0.98**	**71.11±0.18**	50.38±0.63	**46.79±0.94**	**52.62±0.92**	**50.12±0.27**	**36.32±0.82**	**43.77±1.02**
Resnet-18	FGSM	45.54±0.00	67.32±0.00	48.57±0.00	38.39±0.00	45.71±0.00	47.32±0.00	33.04±0.00	41.49±0.10
	I-FGSM	47.08±0.27	69.29±0.31	48.87±0.27	42.26±0.41	50.12±0.10	49.40±0.27	33.75±0.36	42.14±0.36
	SAE	47.62±0.55	69.94±0.37	51.07±0.18	39.17±0.90	48.15±0.27	47.62±0.27	34.88±0.21	42.20±0.27
	ACE	48.39±0.00	69.64±0.00	50.71±0.00	39.64±0.00	48.39±0.00	46.79±0.00	35.54±0.00	42.32±0.00
	AI-FGTM	47.86±0.00	70.89±0.00	51.25±0.00	41.07±0.00	50.18±0.00	48.21±0.00	35.71±0.00	43.21±0.00
	DTA	47.50±0.00	70.71±0.00	51.07±0.00	40.30±0.10	49.11±0.18	47.26±0.10	35.71±0.00	43.21±0.18
	BFA	48.75±0.18	70.83±0.10	51.55±0.21	41.37±0.37	50.00±0.18	47.80±0.10	35.83±0.10	43.04±0.00
	DeCoWA	48.39±0.00	70.71±0.00	**51.79±0.00**	41.55±0.10	50.65±0.10	47.74±0.27	35.71±0.18	43.87±0.21
	SFCoT	46.43±0.11	67.68±0.21	49.11±0.25	40.36±0.19	47.14±0.24	46.96±0.20	33.57±0.14	41.61±0.17
	LADA	**52.56±0.37**	**71.35±0.98**	50.51±0.72	**48.04±0.31**	**54.35±0.37**	**52.20±0.27**	**35.90±0.27**	**45.13±0.21**
Densenet121	FGSM	45.36±0.00	67.50±0.00	48.57±0.00	38.45±0.10	45.83±0.10	47.32±0.00	33.04±0.00	41.43±0.00
	I-FGSM	46.67±0.27	69.29±0.18	48.39±0.31	42.62±0.84	48.51±0.55	51.01±0.10	33.81±0.27	42.02±0.21
	SAE	47.74±0.10	70.30±0.21	50.89±0.36	38.93±0.00	48.21±0.47	47.50±0.31	34.82±0.18	41.67±0.10
	ACE	47.86±0.00	69.29±0.00	50.71±0.00	39.11±0.00	48.39±0.00	46.61±0.00	35.00±0.00	42.50±0.00
	AI-FGTM	48.39±0.00	70.54±0.00	51.25±0.00	40.06±0.10	49.82±0.00	48.21±0.00	35.71±0.00	42.92±0.10
	DTA	47.68±0.00	70.24±0.10	50.95±0.10	39.94±0.27	49.17±0.10	47.26±0.10	35.54±0.00	42.74±0.10
	BFA	48.51±0.21	70.65±0.10	51.49±0.10	42.44±0.27	50.83±0.27	47.74±0.21	35.83±0.10	43.75±0.00
	DeCoWA	48.45±0.21	70.71±0.47	**51.67±0.10**	41.85±0.37	50.06±0.27	47.68±0.00	35.77±0.10	43.87±0.10
	SFCoT	46.25±0.20	68.93±0.25	49.64±0.28	40.36±0.21	47.86±0.27	47.32±0.14	33.57±0.14	42.14±0.28
	LADA	**51.96±0.36**	**70.87±0.37**	50.32±0.47	**47.92±0.98**	**53.57±0.94**	**51.25±0.82**	**36.43±1.07**	**45.95±0.52**
Regnet_x_400mf	FGSM	45.36±0.00	67.50±0.00	48.57±0.00	38.45±0.10	45.71±0.00	47.38±0.10	33.04±0.00	41.43±0.00
	I-FGSM	45.48±0.45	67.74±0.10	47.80±0.41	38.33±0.27	47.98±0.21	49.46±0.71	32.80±0.41	39.58±0.10
	SAE	47.62±0.55	70.36±0.31	50.89±0.31	39.46±0.18	48.04±0.31	47.68±0.00	34.70±0.21	42.08±0.21
	ACE	47.68±0.00	69.58±0.10	50.54±0.00	39.64±0.00	49.29±0.00	47.14±0.00	34.64±0.00	42.68±0.00
	AI-FGTM	47.98±0.10	70.18±0.00	50.71±0.00	39.82±0.00	48.99±0.10	47.86±0.00	35.71±0.00	42.38±0.10
	DTA	47.98±0.27	69.94±0.10	50.65±0.10	39.94±0.10	49.64±0.00	47.14±0.18	**35.77±0.10**	42.38±0.10
	BFA	47.32±0.00	69.64±0.00	50.77±0.10	40.18±0.00	49.29±0.00	47.14±0.00	35.71±0.00	42.32±0.00
	DeCoWA	48.21±0.18	70.24±0.21	**51.25±0.00**	41.25±0.31	49.88±0.10	47.26±0.10	35.60±0.10	43.10±0.10
	SFCoT	45.89±0.28	67.50±0.06	48.75±0.25	39.28±0.27	47.18±0.28	47.50±0.18	33.75±0.30	41.25±0.24
	LADA	**51.73±0.10**	**70.75±0.47**	50.31±0.92	**47.02±0.57**	**52.68±0.47**	**50.77±0.84**	34.49±0.37	**44.18±0.47**

of the image to achieve misjudgment. SAE transforms RGB images into HSV space, and changes hue and saturation components, generating adversarial samples. ACE obtains a color filter based on gradient information, which well balances the attack effect and perceptual quality of the image. These two methods achieve good transferability by adding perturbations only related to the image itself. However, the visual effect of the adversarial samples will be affected. BFA [64] and DeCoWA [65] are two advanced transferable methods. BFA divides the features of images into white box features and black-box features, and obtains the fit image by destroying the white box features to guide the destruction of the black-box features. DeCoWA adopts an elastic deformation to enhance the input samples to obtain rich local details and employs an adaptive control strategy to further constrain the strength and direction of the warping transformation. Table III reports attack results on the RSSCN7 dataset. When attacking ResNet-101 using AlexNet, ResNet-18, DenseNet-121, or RegNetX-400 as surrogates, transferability is limited by significant architectural and capacity gaps. In addition, the self-attention mechanism in ViT prioritizes global semantics and low-frequency components, which mismatches CNN-based perturbations and yields marginal ASR gains. LADA obtains the best attack effect and superior transferability by optimizing perturbations at multiple scales in the latent space. The combination of text embedding and latent representation ensures that adversarial samples retain the original semantics so that the perturbation is difficult to perceive.

2) *Qualitative Results:* Some adversarial samples on AID, NWPU-RESISC, and RSSCN7 generated by LADA and other nine attack methods using Reg-400 as the surrogate model are shown in Figs. 5–7. The perturbation constraint used in this article is 0.1, so the difference

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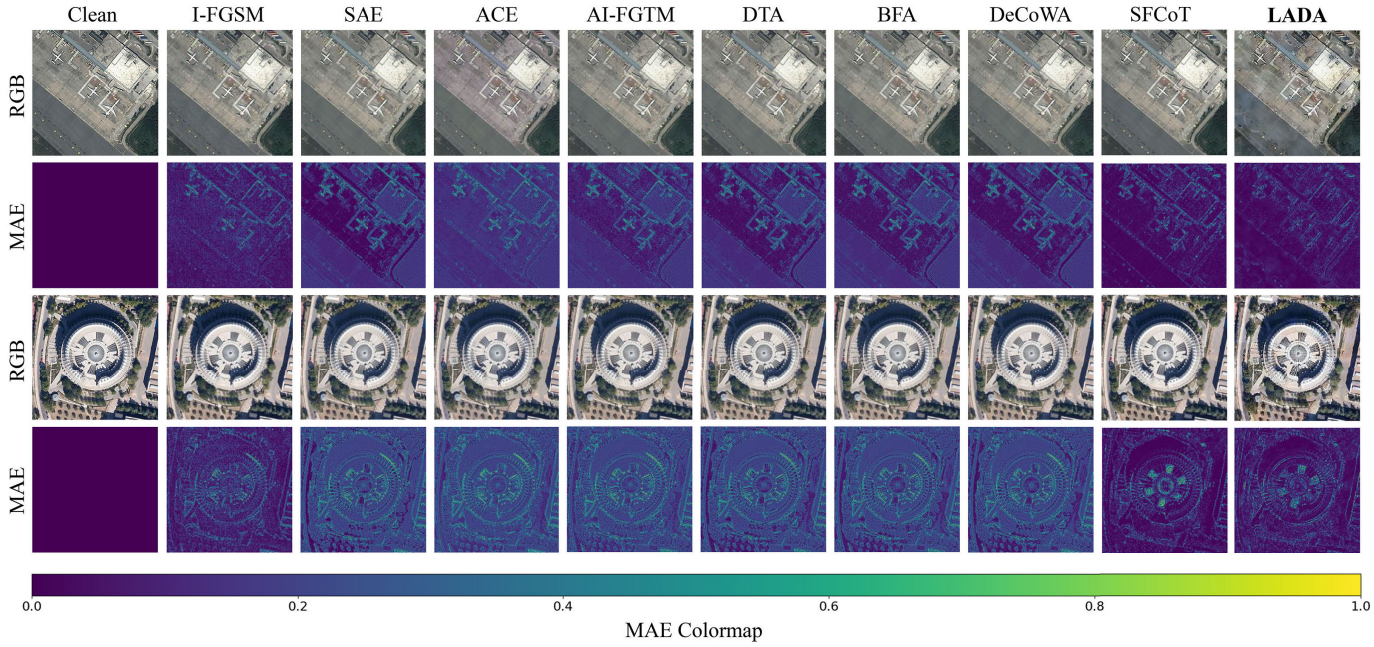


Fig. 5. Adversarial samples generated by nine black-box attack methods using Reg-400 as the surrogate model on AID.

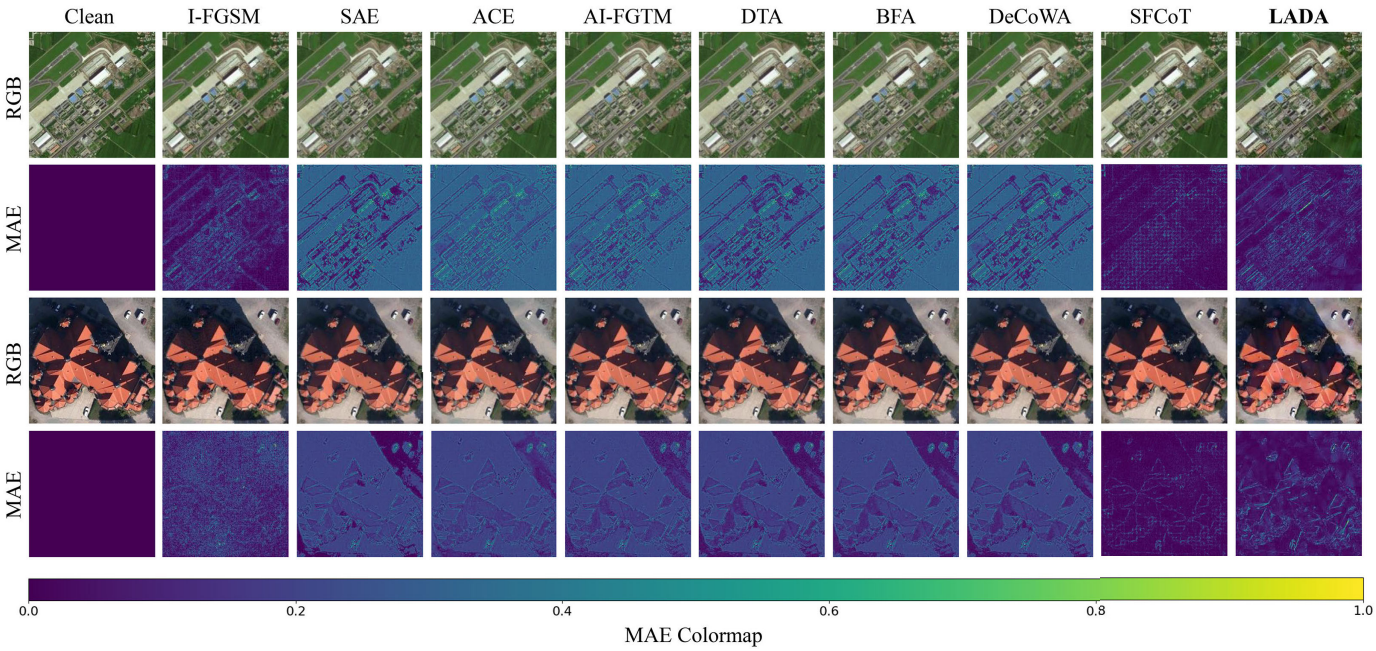


Fig. 6. Adversarial samples generated by nine black-box attack methods using Reg-400 as the surrogate model on NWPU-RESISC.

between the clean images and the adversarial samples is small. To more intuitively show the differences between different adversarial samples, the mean absolute error (MAE) between the generated adversarial samples and clean images is visualized and mapped on the colormap, where darker colors mean smaller differences. From the results, the adversarial samples generated by FGSM, I-FGSM, and SFCoT have a higher similarity with the clean images, but the lower ASRs indicate that the transferability is affected. SAE and ACE perturb the color of clean images so that the adversarial samples have a greater difference. Moreover, the adversarial samples generated by AI-FGSM, DTA, BFA, and DeCoWA are quite different from clean images in texture details and edge parts. Compared with the above methods, LADA optimizes the perturbation in the latent space with the objective function of guiding model misclassification and improving stealthiness, which ensures the transferability of the attack and the imperceptibility of the perturbation.

C. Quality Assessment

This study aims to improve ASR by the black-box attack method and ensure the quality of adversarial samples as much as possible. Therefore, this article adopts three similarity metrics to evaluate the quality of adversarial samples obtained

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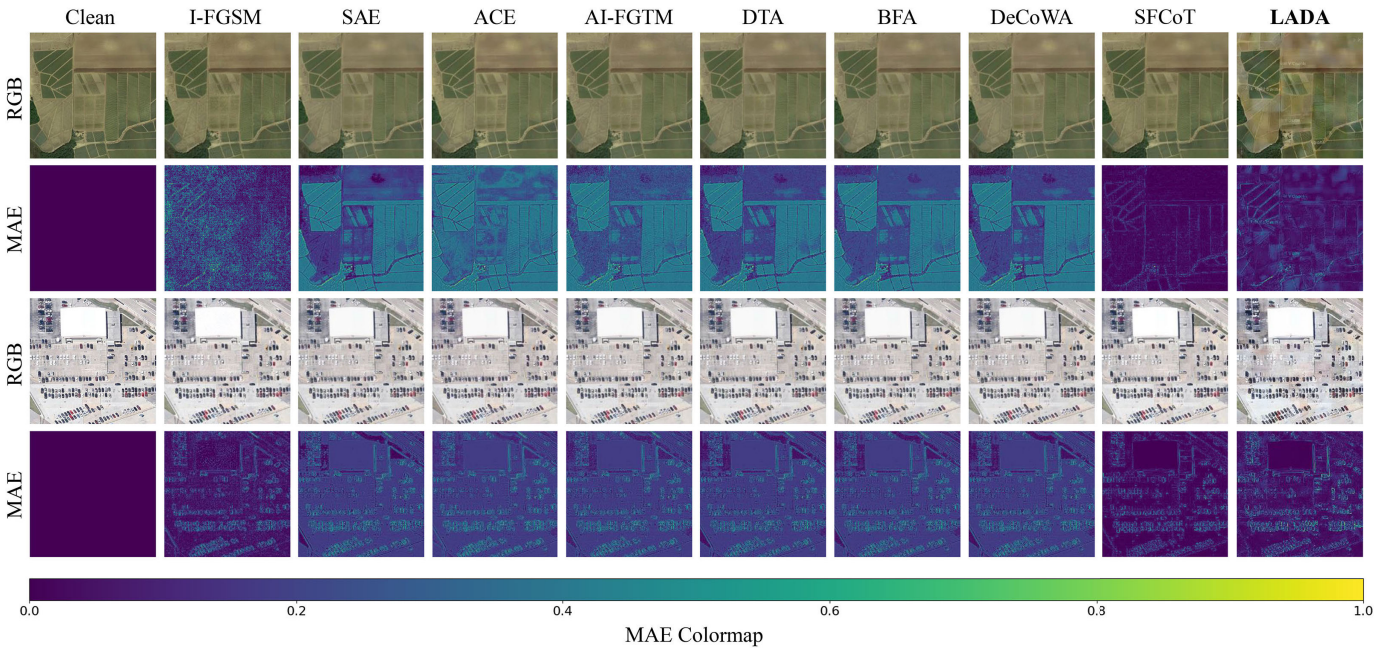


Fig. 7. Adversarial samples generated by nine black-box attack methods using Reg-400 as the surrogate model on RSSCN7.

TABLE IV

COMPARISON OF PSNR, SSIM, AND LPIPS METRICS FOR ADVERSARIAL SAMPLES PRODUCED BY NINE BLACK-BOX ATTACKS WITH RESNET-18 AS THE SURROGATE. BEST VALUES ARE SHOWN IN BOLD

Dataset	Metric	FGSM	I-FGSM	SAE	ACE	AI-FGTM	DTA	BFA	DeCoWA	SFCoT	LADA
AID	PSNR↑	**42.7396**	39.0953	20.5502	20.3073	18.0623	20.7373	20.7026	20.7401	37.9015	23.7463
	SSIM↑	**0.9891**	0.9394	blue0.7974	0.7882	0.5232	0.7946	0.7923	0.7951	0.9103	0.6955
	LPIPS↓	**0.0060**	0.0264	0.0911	0.0940	0.2065	0.0907	0.0906	0.0910	0.0311	0.0704
NWPU-RESISC	PSNR↑	**44.0333**	42.4995	19.5612	19.3941	17.7291	19.6320	19.6168	19.6368	40.2973	24.4671
	SSIM↑	**0.9919**	0.9725	0.7880	0.7803	0.5656	0.7855	0.7834	0.7861	0.9571	0.7360
	LPIPS↓	**0.0036**	0.0144	0.0902	0.0920	0.1756	0.0899	0.0900	0.0900	0.0202	0.0839
RSSCN7	PSNR↑	**42.0399**	38.2387	18.7177	18.5752	17.2194	18.7692	18.7492	18.7740	36.2089	23.7304
	SSIM↑	**0.9880**	0.9359	0.7493	0.7438	0.6011	0.7486	0.7477	0.7492	0.9197	0.6686
	LPIPS↓	**0.0071**	0.0318	0.1004	0.1024	0.1557	0.0985	0.0983	0.0988	0.0210	0.0754

by different attack methods, which are PSNR, SSIM [34], and LPIPS [35]. Table IV shows the PSNR, SSIM, and LPIPS values of the adversarial samples generated by the nine attack methods using Res-18 as the surrogate model. Although the linear perturbations added by FGSM, I-FGSM, and SFCoT have the lowest impact on image quality, their ASR does not have an advantage. LADA ranks third in PSNR and LPIPS, which indicates that the pixel-level difference and perceptual difference between the adversarial samples generated by LADA and the clean image are small. LADA has lower SSIM values, which means that LADA mainly misleads the classifier by changing the structure of images.

D. Ablation Study

This study proposes a multi-scale iterative CMA-ES optimization framework to generate latent-space perturbations that effectively induce misclassification in target models. To evaluate the efficacy of the methodology, we design three experimental variants for ablation analysis. Variant-A: Direct decoding of the initial latent code without perturbation δ; Variant-B: $\bar{z}_0$ with a constrained random perturbation; Variant-C: Single-scale CMA-ES optimization (frequency parameter $f = 8$) applied to Variant-B perturbations. Using DenseNet-121 as the surrogate model on the RSSCN7 dataset, these variants are benchmarked against the proposed LADA method, with comparative results illustrated in Fig. 8.

Variant-A achieves baseline ASR values between 15%–18% without perturbations, which can be attributed to two factors: 1) inherent semantic information encoded within latent representations and 2) architectural discrepancies between surrogate and target models. Variant-C demonstrates 5% higher ASR compared to Variant-B, suggesting that CMA-ES-based optimization enables more effective perturbation construction, albeit showing diminishing returns under single-scale operation. The iterative multi-scale implementation of LADA achieved 10% greater ASR than Variant-C, revealing two critical advantages of our approach: 1) multiscale coordination captures complementary feature-space interactions and 2) iterative refinement overcomes local optima limitations in high-dimensional search spaces. These experimental results confirm that the proposed multiscale CMA-ES

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TABLE V
OA(%) OF THE ADVERSARIAL EXAMPLES GENERATED BY NINE ATTACK METHODS ON DIFFERENT DEFENSE METHODS

Dataset	Defense Methods	FGSM	I-FGSM	SAE	ACE	AI-FGTM	DTA	BFA	DeCoWA	SFCoT	LADA
AID	RS	10.97	12.29	9.53	9.77	13.15	**9.31**	9.37	9.45	9.89	9.36
	NRP	16.98	19.12	15.48	13.74	13.92	16.13	15.84	16.87	17.13	**10.64**
	SR	1.25	1.30	0.52	0.72	0.40	0.61	0.73	0.61	1.11	**0.31**
NWPU-RESISC	RS	12.68	11.88	10.05	10.27	8.75	9.68	9.36	9.61	10.92	**8.44**
	NRP	16.10	15.83	13.55	12.99	13.02	16.15	14.97	14.23	15.07	**12.77**
	SR	0.23	0.43	0.98	0.52	0.46	0.68	0.81	0.34	0.64	**0.11**
RSSCN7	RS	12.17	15.99	10.08	10.57	11.79	10.81	9.89	10.42	14.31	**9.38**
	NRP	27.61	25.39	24.51	24.29	23.95	26.88	26.82	26.95	24.17	**20.16**
	SR	1.81	1.90	1.53	1.59	3.80	**1.47**	1.85	1.50	1.74	1.60

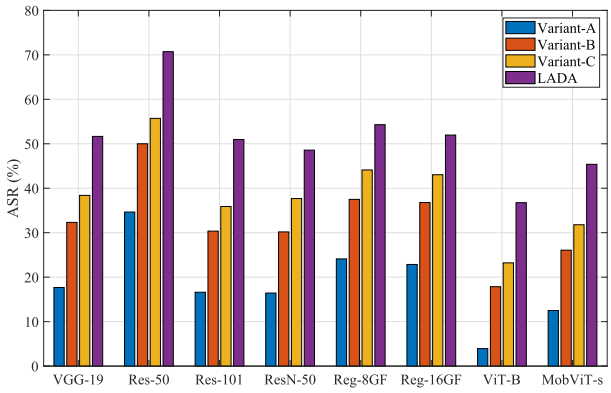


Fig. 8. ASR(%) of LADA and three variants on the RSSCN7 dataset, where the horizontal axis represents the target model.

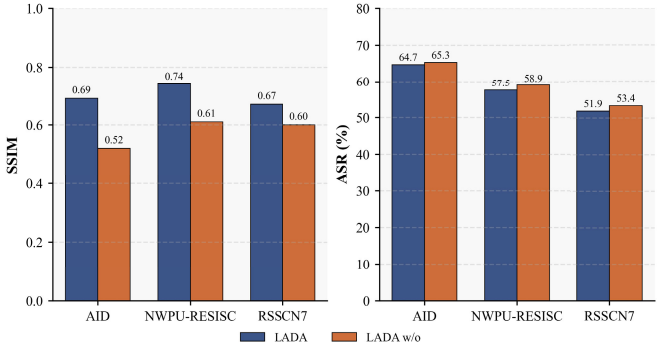


Fig. 9. Comparison between LADA and LADA-w/o in terms of (left) SSIM and (right) ASR, highlighting the effect of text-guided conditioning.

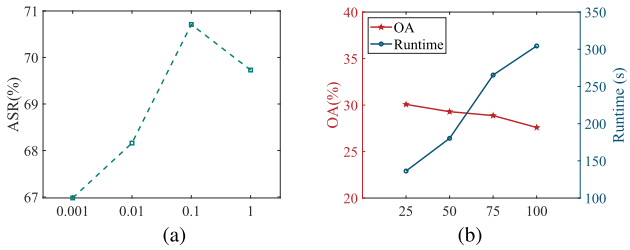


Fig. 10. Effect of different (a) ξ and (b) maxIter on the performance of LADA.

framework significantly enhances adversarial transferability through systematic perturbation space exploration, outperforming conventional single-scale optimization strategies.

Additionally, we investigate the impact of text-conditioned guidance on both the ASR and visual quality of the generated adversarial examples. We compare the LADA framework with a variant that disables the null-text inversion mechanism, referred to as LADA w/o. Fig. 9 shows that LADA consistently achieves higher SSIM scores across all datasets compared to LADA-w/o. This indicates that incorporating null-text inversion can help preserve the structural and perceptual quality of the adversarial examples. LADA-w/o yields slightly higher ASR than LADA. The absence of null-text inversion allows LADA-w/o to freely exploit vulnerable patterns in the image space, leading to marginally higher attack effectiveness. However, this comes at the cost of significantly reduced image quality. These results suggest a trade-off between attack strength and perceptual realism, where null-text conditioning in LADA effectively improves visual quality without substantially compromising the ASR.

E. Results on Defense Methods

To explore the robustness of the proposed LADA, we evaluate the performance of the adversarial samples on the defense methods. Specifically, LADA and nine attack methods utilize Res-18 as the surrogate model to generate adversarial samples, and we select the samples that are misclassified by the target model (Res-50). Subsequently, three common defense methods, RS [66], NRP [67], and SR [68], are tested on these selected samples. The overall accuracy (OA) of the adversarial examples under different defense methods is shown in Table V, where the smaller OA value indicates that the corresponding attack method is sufficient to break through the defense and has stronger robustness. Compared with other attack methods, the adversarial samples generated by LADA on the three datasets maintain lower OA values when facing different defense methods.

F. Parameter Analysis

In LADA, the optimal step size ξ and the maximum number of iterations maxIter of CMA-ES is crucial. To explore their impacts on LADA performance, ξ, and maxIter are set to different values, and experiments are performed on RSSCN7, where Dense-121 and Res-50 are utilized as the surrogate model and the target model, respectively. ξ controls the search range of CMA-ES. The larger ξ can accelerate the convergence speed, and the smaller ξ is helpful to the fine search process. As shown in Fig. 10(a), when ξ is equal to 0.1, LADA has

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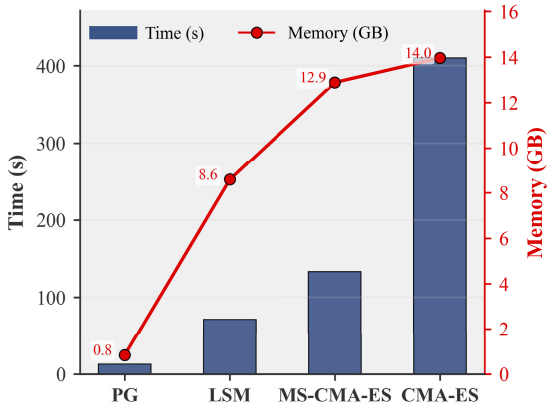


Fig. 11. Componentwise evaluation of runtime and peak memory in LADA framework. "PG:" prompt generation, "LSM:" latent space mapping, "MS-CMA-ES:" multiscale CMA-ES, and "CMA-ES:" single-scale CMA-ES.

the highest ASR and the best performance. Fig. 10(b) shows the OA and runtime of generating adversarial samples under different maxIter. It can be found that the larger maxIter can make the obtained perturbation better and OA smaller, but this will significantly increase the runtime. Considering the variation of runtime and OA, it is recommended that maxIter be set to 50.

G. Complexity Analysis

We analyze the runtime and memory consumption of the LADA framework when attacking a single image. Fig. 11 provides a detailed componentwise evaluation of both computational time and peak memory usage throughout the pipeline. The first stage, prompt generation (PG), incurs the least computational overhead in both time and memory, as we utilize a cloud-based API to obtain textual prompts remotely. The latent space mapping (LSM) component requires DDIM sampling to extract latent codes from the input image, which involves loading a pretrained DM and thus results in higher runtime and memory usage. The core adversarial optimization is performed by the multiscale CMA-ES (MS-CMA-ES), which iteratively generates imperceptible adversarial examples. Specifically, each scale is allowed a maximum of 50 iterations. The multiscale design enables more efficient extraction of salient features at different resolutions, which accelerates convergence by guiding the search direction more effectively. In contrast, the single-scale CMA-ES variant operates at a fixed resolution of 64×64 and requires significantly more iterations to locate viable attack regions and directions due to the absence of multiscale cues. This results in higher computational time and memory consumption.

VI. Conclusion

This article has proposed a LADA framework for generating highly stealthy and transferable adversarial RSI samples. LADA has leveraged denoising diffusion implicit models to map clean RSIs into a latent space, while utilizing DeepSeek-R1 to generate precise captions, ensuring both high efficiency and semantic consistency in adversarial attacks. The integration of multiscale iterative optimization based on CMA-ES has enhanced the search efficiency for globally optimal perturbations. Extensive comparative experiments and quality analyses have been conducted across three benchmark RSI datasets, 14 diverse model architectures, and nine baseline methods, validating that LADA has effectively balanced stealthiness and transferability in black-box adversarial attacks. The generated RSI samples exhibit natural-looking characteristics, yet they have been shown to significantly mislead a wide range of DNNs, including those equipped with defense mechanisms. This capability has inevitably raised significant security concerns for RSI interpretation systems.

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